

Sentiment Classification of Movie Reviews Using Bidirectional LSTM & NLP

TensorFlow / Keras

Bidirectional LSTM

IMDb Dataset

NLP Pipeline

Binary Classification

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SECTION

CSE – A

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01 PROBLEM STATEMENT

- Digital content such as movie reviews, social media posts, and product feedback is growing rapidly.
- Automated systems are needed to understand human emotion and opinion in text.
- Manually classifying thousands of reviews is impractical and time-consuming.
- This project focuses on **binary sentiment classification** of movie reviews.
- The goal is to predict whether a review expresses **Positive** or **Negative** sentiment using a deep learning NLP pipeline.

"Can a Bidirectional LSTM model, trained end-to-end on raw text, learn the nuanced linguistic patterns that distinguish praise from criticism in movie reviews?"

02 DATASET

IMDb Large Movie Review Dataset

50K

Labeled Reviews

50/50

Pos / Neg Balance

233

Avg. Words/Review

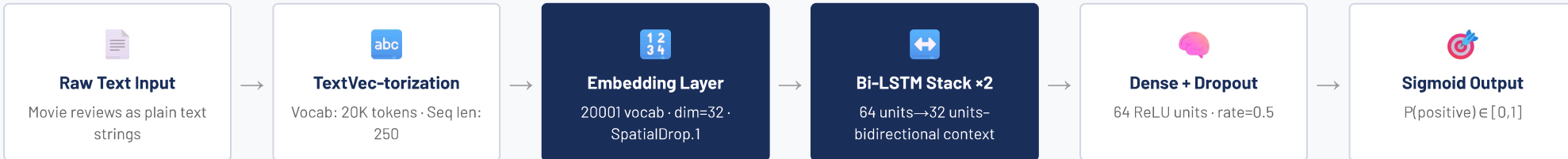
2,470

Max Review Length

● Train 20K ● Test 25K ● Val 5K

30.2% of reviews exceed the 250-token sequence limit; shorter reviews are zero-padded to maintain uniform input dimensions.

03 METHODOLOGY & NLP PIPELINE



04 MODEL ARCHITECTURE

Embedding	(None, 250, 32)	Maps each token to a 32-dim dense vector
↓		
SpatialDropout1D	(None, 250, 32)	Drops entire embedding feature maps (rate=0.3)
↓		
Bi-LSTM (64)	(None, 250, 128)	return_sequences=True – captures full context
↓		
Bi-LSTM (32)	(None, 64)	Condenses sequence into one context vector
↓		
Dense (64, ReLU)	(None, 64)	Non-linear feature transformation
↓		
Dropout (0.5)	(None, 64)	Prevents co-adaptation during training
↓		
Dense (Sigmoid)	(None, 1)	Binary probability output

05 TRAINING CONFIG

OPTIMIZER	LR
Adam	0.001
LOSS	BATCH
BinaryCE	32
MAX EPOCHS	PATIENCE
20	5
VOCAB SIZE	EMBED DIM
20,000	32

Early Stopping monitors val_loss and restores best weights automatically to prevent overfitting.

APPROACH COMPARISON

Approach	IMDb Acc.
Lexicon-based	~72%
ML(SVM/NB)	~83%
Unidir. LSTM	~89%
BiLSTM (ours)	83.5%
BERT / Transformer	~94%

06 RESULTS & DISCUSSION

83.5%

TEST ACCURACY

0.3965

TEST LOSS

Class	Prec.	Recall	F1
NEG	0.84	0.83	0.83
POS	0.83	0.84	0.84
Macro Avg	0.84	0.84	0.83

Balanced precision & recall across both classes confirms the model is unbiased – a direct result of the perfectly balanced 50/50 dataset.

KEY FINDINGS & DISCUSSION

✔ **Strong balanced performance:** F1-scores of 0.83–0.84 for both classes show the model is well-calibrated and does not favour either sentiment.

✔ **High-confidence predictions:** Clearly positive reviews (e.g., "One of the best movies I've ever seen") scored >90% confidence; clearly negative reviews scored similarly high.

⚠ **Double-negation misclassification:** The phrase "not bad" was labelled NEGATIVE (73.2% confidence) because the model associated "not" with negative contexts during training.

⚠ **Mild overfitting:** Training accuracy of 88–90% vs. test accuracy of 83.5% – managed effectively by EarlyStopping.

→ **Future Work:** Replace learned embeddings with GloVe/FastText; fine-tune BERT for expected ~93% accuracy; implement dedicated negation preprocessing.